

Lab 02 Arrays and File I-O

- Q1 Fixing a simple program
- Q2 Multiplication table (printf, nested loops)
- Q3-4 File reading + 2D array
- Q5 An Attendance Checking Program (201617A Quiz 1)
- Q6 [Optional] Multiplication table with a frame

Q1. The given Java program contains a number of syntax and logical errors. Fix ALL of them.
(The purpose of the program is given as comments in the code.)

```
public class Main {
    public static final int WIDTH = 12;
    public static final int HEIGHT = 8;

    // Given a 2D array of integers, count the values which are between 0 and 100 inclusive.
    // The size of the array is a WIDTH x HEIGHT. The array is passed as the parameter A.
    public static void count_0_to_100(int[HEIGHT][WIDTH] A) {
        int count = 0;
        for(int y = 0, y < HEIGHT, y+1) {
            for (int x = 0, x < WIDTH, x+1)
                if(0 <= A[y][x] <= 100)
                    count+1;
        }
        System.out.print("The count is: " count);
    }

    public static void main(String[] args) {
        int[HEIGHT][WIDTH] values;
        int x, y;

        //create some data in values[][]
        for(y = 0 ; y < HEIGHT; y++)
            for(x = 0 ; x < WIDTH; x++)
                values[y][x] = ((x+13)*(y+29))%413-139;

        //display contents in values[][]
        for(y = 0 ; y < HEIGHT; y++) {
            for(x = 0 ; x < WIDTH ; x++)
                System.out.printf("%5d", values[y][x]);
            System.out.println();
        }

        //call the function count_0_to_100 for counting(0...100)
        count_0_to_100(values[HEIGHT][WIDTH]);
    }
}
```

Guidelines:

- There are at least 10 errors.
- Avoid *Trial-and-error* !! It often wastes time and brings no advantage for learning.
- For the `count_0_to_100()` method, the array parameter should be: `int[][] A`.
- For `main()`, the 2D array should be created as `int[][] values = new int[HEIGHT][WIDTH];`
- In no more than 5 minutes, fix all remaining problems. The program should run as shown below.

To weaker students: Please directly read the *Hints 1 to 10*.

Then redo after class to practice and study the details again (100% digestion).

Expected output (No user input is needed):

238	267	-117	-88	-59	-30	-1	28	57	86	115	144
251	-132	-102	-72	-42	-12	18	48	78	108	138	168
264	-118	-87	-56	-25	6	37	68	99	130	161	192
-136	-104	-72	-40	-8	24	56	88	120	152	184	216
-123	-90	-57	-24	9	42	75	108	141	174	207	240
-110	-76	-42	-8	26	60	94	128	162	196	230	264
-97	-62	-27	8	43	78	113	148	183	218	253	-125
-84	-48	-12	24	60	96	132	168	204	240	-137	-101

The count is: 25

- Test the program and submit to PASS (only 1 test case: no input)

Now, review the program. Make sure you understand the code.

Q2. Complete a program that prints out **a multiplication table** as shown below:

Sample output (Underlined content is input by the user):

```
Input the width of the multiplication table (2-10): 9
Input the height of the multiplication table (2-10): 9
1| 2| 3| 4| 5| 6| 7| 8| 9|
2| 4| 6| 8| 10| 12| 14| 16| 18|
3| 6| 9| 12| 15| 18| 21| 24| 27|
4| 8| 12| 16| 20| 24| 28| 32| 36|
5| 10| 15| 20| 25| 30| 35| 40| 45|
6| 12| 18| 24| 30| 36| 42| 48| 54|
7| 14| 21| 28| 35| 42| 49| 56| 63|
8| 16| 24| 32| 40| 48| 56| 64| 72|
9| 18| 27| 36| 45| 54| 63| 72| 81|
```

Note:

1. Download the given program framework.

```
public class Main {
    public static void main(String[] args) {
        int width, height;

        System.out.print("Input the width of the multiplication table (2-10): ");
        System.out.print("Input the height of the multiplication table (2-10): ");

        /* Notes to students:
        1. We need a Scanner object to get user input (like Lab01-Q1 in Lab01.pdf Page 3):
        - Add the import statement at the beginning: import java.util.*;
        - Declare a scanner object: Scanner [object variable name];
        - Create a scanner object as: new Scanner(System.in);
        - To read an integer, we call the .nextInt() method: [scanner object].nextInt();
        - Close the scanner object: [scanner object].close();

        2. Learn from Q1: System.out.printf("%5d", x); // Show x, "%" means a field,
                                                    // right aligned, width=5,
                                                    // d means decimal

        */

    }
}
```

2. Do NOT use any array.
3. No need to check user input.
4. Learn from Q1: `System.out.printf("%5d", x);` ← Show x, "%" means a field, right aligned, width=5, d means decimal
5. To output the vertical stroke `|` character, `System.out.print("|");` Vertical stroke
6. To output a line break: `System.out.println();`



Submit the program to PASS.

**** Make sure you get 100% correct in PASS for Q1 and Q2.**

Q3. You are given a program to finish (Main.java, Table2dMxSumRowCol.java), and data files (1.txt, 2.txt etc).

The program performs these actions:

- inputs integers from a data file and puts them in a 10x10 2-D array,
- prints the 2D array
- summarizes the maximum sum among the rows as well as the columns.

Sample output (Underlined content is input by the user):

```
Please input the file pathname: c:\1.txt
0 0 0 36 0 0 0 0 0 0
0 0 0 0 0 0 0 3 0 29
0 0 14 0 0 0 0 0 0 4
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 21 0 0 0
0 0 0 0 0 0 9 0 0 0
0 0 0 0 0 30 0 0 0 18
0 0 0 0 45 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
39 0 37 0 0 0 0 0 0 0
Maximum row sum: 76
Maximum col sum: 51
```

1.txt - Notepad

File	Edit	Format
2	2	14
9	2	25
2	9	4
6	9	18
0	3	36
5	6	9
6	5	30
9	0	39
9	2	37
1	9	29
7	4	45
1	7	3
4	6	21

Data file:

Each line in the data file has a *row number*, then a *column number*, and then the *value* at that location.

E.g., 2 2 14 means to put 14 at column 2, row 2.

Details:

- About Table2dMxSumRowCol.java:
 - The Table2dMxSumRowCol class encapsulates the 2D array.
 - Its constructor reads the data file and put the data into the 2D array.
 - It also provides the following *public methods*:

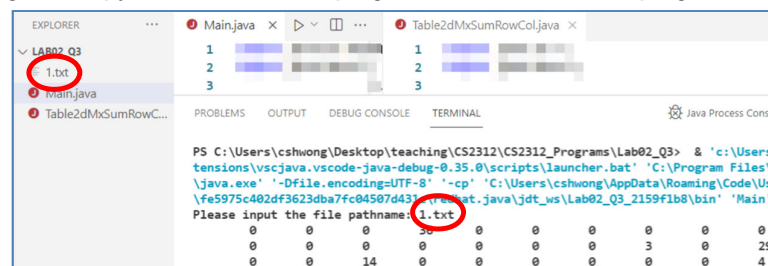

```
int getColSumMax()
int getRowSumMax()
void print() //print the 2D array
```
- The 2D array should have all zeros at the beginning. (This is done in Java by default, no worries.)
- Values extracted from the data file should be put into the 2D array one by one. If the same location appears in the file 2 times or more, the later one should override the previous one(s).

Guidelines:

Step 1. **First**, finish the required code in Main.java.

Step 2. Open Given_Code.txt. Please make use of the given code for Table2dMxSumRowCol to finish the 2D array, constructor, .print()
Test the program and make sure print() runs correctly.

To test the program, copy a test case to the program folder and run the program as follows:



Step 3. Complete getRowSumMax(), getColSumMax() in Table2dMxSumRowCol.java

Test the program and submit to PASS.

For guideline and demo, please find the demo video on <https://www.cs.cityu.edu.hk/~helena/cs231220..>

For Troubleshooting, read Canvas => CS2312 => Piazza => *Lab02Q03 Troubleshooting*

- Q4.** Extend your program in Q3 such that it also displays the row and column numbers (0..9). In case the maximum sum appears in more than one row, or more than one column, output all of them, separated by a comma.

```

Please input the file pathname: c:\2.txt
0 0 0 36 0 0 0 0 0 0
0 0 0 0 0 0 0 3 0 29
0 0 14 0 0 0 0 0 0 4
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 21 0 0 0
0 0 0 0 0 0 9 0 0 0
0 0 0 0 0 30 0 0 0 18
0 0 31 0 45 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0
39 0 37 0 0 0 0 0 0 0
Maximum row sum: 76 (row 7,9)
Maximum col sum: 82 (col 2)

```

This question will be discussed briefly during the lecture before the deadline. But please try to figure out the solutions on your own first.

Q5 An Attendance Checking Program (201617A Quiz 1)

Finish the program in the quiz paper (Quiz 1 of 2016-17 Semester A) with the given source files. Submit to PASS before the deadline.

Q6. [Optional]

Rewrite your program in Q2 to print out a multiplication table with a frame, as shown below:

Sample output:

```

Input the width of the multiplication table (2-10): 6
Input the height of the multiplication table (2-10): 4
/-----\
| 1|  2|  3|  4|  5|  6|
|-----|
| 2|  4|  6|  8| 10| 12|
| 3|  6|  9| 12| 15| 18|
| 4|  8| 12| 16| 20| 24|
\-----/

```

Note that you print "" as `System.out.print("\\");` (i.e., *Escape sequence* for "\")

This is an optional exercise but is worth doing. It drills your patience and skill.

Have fun! ☺